Neurolinx | Railway Solutions

INTELLIGENT | SAFER | MORE RELIABLE

FogPASS

FOG NAVIGATION & LANDMARK ASSISTANCE SYSTEM FOR RAILWAYS

A smart, in-cab solution that enhances visibility, navigation and landmark awareness, enabling safer and more reliable train operations in fog and low-visibility conditions.



- Accurate Navigation (GPS/GNSS & NavIC)
- Advance Landmark Warnings
- English & Hindi Announcements
- Auto & Manual Route Selection
- Foggy & Non-Foggy Modes
- Upcoming Landmark Ladder

Built for the **locomotive cab**

FogPASS is a railway navigation and landmark-awareness solution designed around the operational realities of locomotive driving. It brings route context, upcoming landmarks and timely announcements into a focused driver-facing experience.



A CLEARER VIEW OF WHAT LIES AHEAD

The system combines satellite-based navigation with route selection, landmark alerts and visibility-specific modes. Its purpose is to help drivers access relevant information in a structured and easy-to-follow format.

PRODUCT FOCUS

Navigation / Landmark awareness / Bilingual audio / Route selection / Fog-mode options

GPS/GNSS & NavIC Navigation

Location awareness that supports route context.



FEATURE 01

FogPASS uses satellite-based navigation information to present the locomotive position and help establish where the train is along its selected route.

KEY CAPABILITIES

Position information

Provides location context in the driver-facing display.

Route orientation

Helps relate current position to the route being followed.

Driver experience

Makes navigation information accessible within the cab.

Advance Landmark **Warnings**

Timely information before approaching route landmarks.

FEATURE 02

Advance landmark warnings are intended to notify the driver about relevant upcoming railway locations, supporting anticipation and situational awareness.



KEY CAPABILITIES

Early awareness

Highlights approaching landmarks before they are reached.

Relevant prompts

Keeps attention on route features that matter to the journey.

Information support

Complements, but does not replace, railway signals and operating instructions.

English & Hindi **Announcements**

Spoken information to complement the screen.



FEATURE 03

FogPASS includes English and Hindi announcements to communicate navigation-related information through an additional channel. Audio guidance can reduce the need to rely on visual information alone.

KEY CAPABILITIES

Bilingual guidance

Offers announcements in English and Hindi.

Audio + visual

Pairs spoken cues with on-screen route information.

Cab-focused design

Supports clear communication without unnecessary complexity.

Auto & Manual
Route Selection

Flexible route setup for different operating needs.

FEATURE 04

Automatic and manual route-selection options allow the system to establish route context in more than one way. The chosen route provides the basis for the upcoming landmark information shown to the driver.



KEY CAPABILITIES

Automatic option

Provides a route-selection workflow intended to reduce manual interaction.

Manual option

Allows route selection when a manual choice is appropriate.

Consistent guidance

Keeps route-related information organised around the selected route.

Foggy & **Non-Foggy Modes**

Purposeful information across visibility conditions.



FEATURE 05

FogPASS provides distinct foggy and non-foggy modes. These modes are designed to support a suitable information experience when the external view and operating environment change.

KEY CAPABILITIES

Foggy mode

Provides a dedicated operating option for low-visibility conditions.

Non-foggy mode

Provides an option for normal visibility conditions.

Operational continuity

Keeps navigation and landmark information available across modes.

Upcoming
Landmark Ladder

A sequential look at what comes next.

FEATURE 06

The upcoming landmark ladder organises approaching route landmarks in a clear sequence. It helps the driver review the order of upcoming locations and relate them to the current journey.



KEY CAPABILITIES

Look-ahead sequence

Presents approaching landmarks in an ordered list.

Route understanding

Supports awareness of the next relevant locations.

Readable layout

Keeps landmark information structured and easy to scan.

One connected **information flow**

FogPASS brings together position, route selection, upcoming landmarks and driver announcements into a single information experience. Each feature contributes a different part of the route-awareness workflow.

01

ESTABLISH ROUTE CONTEXT

GPS/GNSS and NavIC navigation provide position context alongside automatic or manual route selection.

02

SHOW WHAT IS AHEAD

The landmark ladder presents upcoming railway locations in sequence.

03

COMMUNICATE IMPORTANT CUES

Advance landmark warnings and English/Hindi announcements support driver awareness.

04

SELECT VISIBILITY MODE

Foggy and non-foggy modes provide alternative operating views for different conditions.

Designed around **driver awareness**



SIX SMART FUNCTIONAL FEATURES

01 GPS/GNSS & NavIC Navigation

02 Advance Landmark Warnings

03 English & Hindi Announcements

04 Auto & Manual Route Selection

05 Foggy & Non-Foggy Modes

06 Upcoming Landmark Ladder

FogPASS is an informational support system. Applicable railway operating rules, signalling instructions and approved safety procedures remain authoritative.

NEUROLINX

FOGPASS / RAILWAY SOLUTIONS



Smarter route awareness.

Clearer information ahead.

GPS/GNSS & NavIC | Landmark warnings | English & Hindi announcements | Route selection | Foggy & non-foggy modes | Landmark ladder